

PART 1 – Company Profile

Company Information

Company Name: ABC Startup Solutions

Nature of Business: Software Development and IT Solutions

Office Location: Santa Rosa City, Laguna, Philippines (Fictional)

Company Size: 20 Employees

Company Vision

Our mission is to be a top software development firm that provides innovative, reliable, and technology-driven solutions that assist businesses in enhancing productivity and accelerating digital transformation.

Company Mission

ABC Startup Solutions offers top-quality software applications and IT solutions designed to satisfy the needs of modern enterprises. The organization is committed to providing exceptional customer service, innovating, and creating a healthy work environment that fosters continual learning and development of its employees.

Organizational Structure

Chief Executive Officer (CEO)

- Information Technology Department
 - IT Manager
 - System Administrator
 - Software Developers (3)
- Human Resources Department
 - HR Manager
 - HR Staff (3)
- Finance Department
 - Finance Manager
 - Finance Staff (4)
- Sales Department
 - Sales Manager
 - Sales Representatives (5)

Employee Distribution

Department	Number of Employees
Information Technology	5
Human Resources	4
finances	5
Sales	6
Total	20

Business Requirements

ABC Startup Solutions is a freshly formed software development firm that needs a reliable and scalable IT infrastructure to support everyday business operations, software development activities, communication, data storage, internet access, and information security. The infrastructure must be able to accommodate the existing staff of 20 employees and have room for future development as the firm grows.

PART 2 – Enterprise Hardware Inventory

Hardware Infrastructure Overview

ABC Startup Solutions needs solid hardware infrastructure to support its 20 employees and regular business operations. The suggested hardware inventory covers employee PCs, servers, networking equipment, printing devices, backup storage, and power protection equipment. The numbers chosen are based on the number of employees in each department, taking into account future growth and operational reliability.

Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Dekstop Computer	15	IT	Main workstation for employees who work mostly in the office
HW-002	Laptop Computer	5	HR, Finance	Provides mobility for employees attending meetings and presentations, requiring remote access
HW-003	Server	1	IT	Centralized services, file storage, user administration, and other internal services
HW-004	Monitor	20	All Departments	Provides a separate display for each employee

				workstation
HW-005	Router	1	IT	Provides network routing and bridges the internal network to the Internet
HW-006	Managed Network Switch	1	IT	Connects computers, servers, printers, access points, and other wired devices
HW-007	Laser Printer	2	HR, Finance	It provides document printing for administrative and financial activities.
HW-008	UPS	3	IT	Offers interim power and protection for mission-critical IT equipment during power outages
HW-009	Wireless Access Point	2	IT/Office-wide	Provides wireless network access for laptops and approved mobile devices
HW-010	NAS Storage	1	IT	Provides central storage on the network and a secondary site for backup data
HW-011	External Backup Drive	2	IT	Provides offline backup storage for important company data
HW-012	Server Rack	1	IT	Provides secure and organized physical storage for server and network equipment

Hardware Justification

Desktop Computers

For employees who work mostly in the office, it is advised to have 15 desktop PCs. Desktop computers are often more affordable and easier to maintain than equipping all employees with laptops and provide reliable service for routine corporate functions.

Laptop Computers

Five laptops are recommended for employees who require mobility. These can be assigned primarily to selected IT and sales personnel who may need to attend meetings, presentations, client discussions, or perform work outside their normal workstation.

Server

A dedicated server is advised to provide consolidated IT services. The server can host internal applications, file sharing services, user management services, and other resources needed by the organization. Leave the server management to the IT department.

Monitors

It is advisable to have twenty monitors so that each employee has their own dedicated screen. Depending on workload and technological needs, IT professionals may add additional displays in the future.

Router and Managed Switch

For dependable network connectivity, we propose one business-class router and one managed network switch. The router will manage traffic between the internal network and the internet, while the managed switch will connect wired devices throughout the company.

Printers

Two network laser printers are recommended. One printer can be mostly for the HR department, and the other can be for finance and other departments. This is what network printing means. Each department doesn't have to have its own printer.

UPS

Three UPS units are suggested to protect the life-supporting infrastructure from unforeseen power outages and voltage fluctuations. Focus on critical IT devices such as servers, network equipment, and others.

Wireless Access Points

We propose 2 wireless access points to provide sufficient wireless coverage at the workplace level. You have a better chance of avoiding network congestion, and a single access point can cover a larger area than several access points.

NAS Storage

Centralized file storage and backup with a single NAS device. The NAS can be used as an additional storage location for key company files and to support the organization's backup plan.

External Backup Drives

We recommend using two external backup disks for offline backups. Having backup copies outside the main network adds protection from hardware failure, inadvertent deletion, and some types of cyberattacks.

Server Rack

Using one server rack is recommended for the secure and organized storage of servers, network switches, routers, firewalls, patch panels, and other network equipment. Good physical arrangement facilitates maintenance and troubleshooting.

Hardware Allocation Summary

The proposed hardware architecture provides all of the 20 employees with a primary computing device, as well as centralized servers, network connectivity, printing, storage, and power protection. The design does not include the purchase of unneeded equipment, but it is sufficient for the existing activities of the organization.

As ABC Startup Solutions grows, you can scale up the infrastructure. You may add more PCs, access points, storage devices, and network equipment without rebuilding the whole infrastructure.

PART 3 – Enterprise Software Inventory

ABC Startup Solutions will use a combination of operating systems, productivity software, development tools, security applications, and system administration utilities to support its daily operations.

Software Inventory

Software	Version	License	Purpose
Window 11 Pro	Current	Commercial	Employee computer operating system
Ubuntu Server	LTS	Open Source	Server operating system for corporate servers
Microsoft 365	Current	Subscription	Documents, spreadsheets, presentations, and productivity
Visual Studio Code	Current	Free	Software development and code editing
Git	Current	Open Source	Source code version control
GitHub Desktop	Current	Free	Graphical Git and GitHub management
VirtualBox	Current	Open Source	Virtual machine creation and testing
Google Chrome	Current	Free	Web browsing and web applications

Microsoft Defender	Current	Included	Antivirus and endpoint protection
AnyDesk	Current	Commercial	Remote IT support and troubleshooting
7-Zip	Current	Open Source	File compression and extraction

Software Justification

- **Windows 11 Pro** – Provides a secure and business-ready operating system for employee computers.
- **Ubuntu Server** – Provides a reliable and cost-effective server operating system.
- **Microsoft 365** – Supports daily office productivity and documentation.
- **Visual Studio Code**—Used by IT personnel for programming and development.
- **Git** – Tracks and manages changes in software source code.
- **GitHub Desktop** – Simplifies Git repository management.
- **VirtualBox**—Allows IT personnel to create virtual machines for testing.
- **Google Chrome** – Provides access to websites and web-based applications.
- **Microsoft Defender**—**Protects** Windows devices from common malware and security threats.
- **AnyDesk**—**Allows** authorized IT personnel to provide remote technical support.
- **7-Zip** – Used to compress and extract files.

Software Management

The IT staff will keep track of the software inventory, install allowed applications, apply security upgrades, and prevent unauthorized software installations. “Most tools for development will be installed on IT workstations, and we will provide productivity and security applications in accordance with employee needs.

PART 4 – Enterprise Network Inventory

ABC Startup Solutions requires a reliable wired and wireless network to connect employees, servers, printers, and other IT equipment. The following networking equipment is recommended for the 20-employee office.

Network Equipment Inventory

Asset ID	Equipment	Quantity	Specification	Purpose
NET-001	ISP Modem/ONT	1	Gigabit-capable	Connects the office to the internet service provider
NET-002	Router	1	Business-class	Manages network routing and internet traffic
NET-003	Firewall	1	Business/Next Generation	Protects the internal network from external threats
NET-004	Managed Switch	1	48-port Gigabit	Connects wired computers, servers, printers, and access points
NET-005	Wireless Access Point	2	WI-FI 6	Provides wireless connectivity throughout the office
NET-006	Patch Panel	1	48-port CAT6	Organizes and manages network cable connections
NET-007	CAT6 Ethernet Cable	500 meters	CAT6	Provides wired network connections
NET-008	RJ45 Connectors	100 pieces	CAT6-compatible	Used to terminate Ethernet cables

Network Design Summary

The Internet connection from the ISP modem will be provided to the firewall. The firewall will ensure network security before the traffic reaches the router and internal network. The company's wired devices will be connected to the controlled switch, and wireless access points will enable Wi-Fi connectivity.

The 48-port switch and patch panel are suggested over smaller equipment to provide additional ports for future employees and devices and network expansion.

The internet connection will come into the ISP modem and then be passed through to the firewall. The firewall will ensure network security before the traffic reaches the router and internal network. The company's wired devices will connect to the controlled switch, and the wireless access points will enable Wi-Fi access.

A 48-port switch and patch panel are recommended in place of lesser equipment to enable additional ports for future employees, devices, and network growth.

PART 5 — Network Diagram

(Already done using Draw.io)

PART 6 – System Administration Roles

System administration is a collaborative effort between multiple IT specialists who work together to maintain an organization's computers, networks, servers, and cloud services.

6.1 Helpdesk Technician

Responsibilities:

- Provides first-level technical support to staff.
- Troubleshoots hardware, software, and account issues.
- Handles and escalates unsolved issues.

Skills:

- Basic hardware and software problem-solving
- Customer Services
- Communication and problem-solving

Common Tools:

- AnyDesk
- Windows tools
- Ticketing systems
- Microsoft 365

Certifications:

- CompTIA A+
 - Google IT Support Professional Certificate
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6.2 Network Administrator

Responsibilities:

- Configures and supports routers, switches, and access points.
- Monitors network performance and connectivity.
- Troubleshoots network and security problems.

Skills:

- TCP/IP networking
- Network troubleshooting
- Router and switch configuration
- Network security

Common Tools:

- Cisco Packet Tracer
- Wireshark
- PuTTY
- Network monitoring tools

Certifications:

- CompTIA Network+
 - Cisco CCNA
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6.3 Linux System Administrator

Responsibilities:

- Manages Linux servers & services.
- Creates user accounts and maintains permissions.
- Performs system upgrades, backups, and troubleshooting.

Skills:

- Linux command line
- Shell scripting
- File permissions
- Server security

Common Tools:

- Ubuntu Server
- SSH
- Bash
- VirtualBox

Certifications:

- Linux Foundation Certified System Administrator (LFCS)
 - Red Hat Certified System Administrator (RHCSA)
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6.4 Cloud Administrator

Responsibilities:

- Manages cloud servers, storage, and network resources.
- Monitors cloud systems and performance.
- Implements cloud security and access controls.

Skills:

- Cloud computing
- Virtualization
- Networking
- Cloud security

Common Tools:

- Microsoft Azure
- Amazon Web Services (AWS)
- Azure Portal
- AWS Management Console

Certifications:

- Microsoft Certified: Azure Administrator Associate
 - AWS Certified Cloud Practitioner
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Collaboration Between IT Professionals

The Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator work collaboratively to maintain reliable IT services. The helpdesk technician troubleshoots employee difficulties and escalates technological issues as needed. The network administrator controls network connectivity and infrastructure. The Linux system administrator manages Linux-based systems and services. Cloud Administrator. The Cloud Admin is responsible for managing the cloud resources and services. Through communication and exchanging information, these experts are able to quickly resolve issues and keep the organization's IT infrastructure secure and available.

PART 7 – Infrastructure Recommendations

The ABC Startup Solutions IT infrastructure must be stable, secure, and scalable to support the present 20 employees and future expansion.

7.1 Internet Provider

The corporation is to have business-grade internet with a speed of at least 300-500 Mbps. A business plan is recommended since it can provide better reliability, technical support, and service options for a corporation.

7.2 Server Specifications

The company should use a dedicated server with the following recommended specifications:

- 8-core processor
- 32 GB ECC RAM
- SSD storage with RAID 1
- Ubuntu Server LTS
- Gigabit Ethernet connectivity

This configuration provides sufficient performance for the company's initial server requirements and allows room for future expansion.

7.3 Backup Strategy

ABC Startup Solutions should follow the **3-2-1 backup strategy**:

- 3 copies of important data
- 2 different storage media
- 1 copy stored off-site

7.4 Security Recommendations

The company should implement the following security measures:

- Business firewall
- Regular operating system and software updates
- Antivirus and endpoint protection
- Multi-factor authentication (MFA)
- User access controls
- Regular data backups
- Employee security awareness training
- Secure Wi-Fi configuration

7.5 Antivirus

Windows 11 Pro machines must have Microsoft Defender turned on and kept up to date. The IT team should monitor endpoint protection and make sure security upgrades are rolled out on time.

7.6 Password Policy

Employees should follow a company password policy that includes:

- Minimum password length of 12 characters
- Unique passwords for company accounts
- MFA for supported services
- No password sharing
- Immediate password changes when an account may be compromised

7.7 Expansion Plan

The infrastructure should be suitable for 20 employees and able to expand to 40-50 employees. The 48-port network switch, additional wireless access points, scalable storage, and upgradeable server resources will enable the organization to grow without replacing the entire infrastructure.

PART 8 – Personal Reflection

On the Enterprise Infrastructure Planning project, I learned that system administration is much more than just keeping computers up and running and resolving technical issues. What I learned is that adequate planning is part of developing a reliable IT infrastructure. Before buying equipment or setting up a network, a system administrator must understand the organization's requirements, budget, number of employees, security demands, and future expansion plans.

I learned how to take hardware, software, and network inventories for this project. I learned that every piece of hardware should have a purpose and that the amount of gear should depend on the real demands of the company. I also learned how multiple IT positions, like helpdesk technicians, network administrators, Linux system administrators, and cloud administrators, work together to keep business operations running.

For me, the hardest part was creating the corporate network diagram. I wanted to know how to connect the internet, the modem, the firewall, the router, the switch, the server, the wireless access point, the departments, and other devices. I created a diagram to help illustrate how different parts of an organization's infrastructure communicate with each other.

Planning is critical at the time of deployment, as it can help you eliminate excessive spending, configuration problems, security concerns, and future scalability challenges. A well-planned infrastructure allows an organization to use its resources efficiently and makes troubleshooting and maintenance easy. It also makes sure the infrastructure can handle the company's existing operations and future development.

This project will assist me in becoming a better system administrator by developing my skills in technical planning, documentation, issue resolution, and communication. It also taught me to conceive of the infrastructure as a whole system and not just individual computers or gadgets. In the future, I will use these talents in building, deploying, maintaining, and enhancing IT systems for real enterprises.